# Next-Day Weather Forecast by Machine Learning Models

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Github Link: https://github.com/weijiawang2003/DATA-1030-Project

## Introduction

Weather follows regular cycles, most clearly the annual progression of the four seasons, making it suitable for modeling with machine learning methods. We aim to develop lightweight models for next-day mean-temperature prediction in Providence, Rhode Island. The models will use recent weather observations, historical weather from the same period in previous years, and calendar information. These predictions may help students and visitors plan daily travel and outdoor activities. We use ERA5 reanalysis data (Hersbach et al., 2020), a widely used and well-documented source.

Several studies have applied machine learning to city-scale weather prediction. Kinast and Fokoué (2023) used approximately 20 years of NOAA station data to compare ARIMA, exponential smoothing, multilayer perceptrons (MLPs), and Gaussian processes across five U.S. cities, including Boston. MLP performed best in their comparison. Jakaria et al. (2018) used random forests to predict next-day hourly temperature in Nashville and found that adding data from neighboring cities improved performance. Han and Randall (2026) used ERA5 data with random forests and GRUs to predict next-day heat index in four U.S. cities. Large-scale systems such as GraphCast (Lam et al., 2023) and Pangu-Weather (Bi et al., 2023) were trained on decades of global weather data and achieved strong forecasting performance, but require substantial data and computing resources for training.

Our project focuses on a single grid point near Brown University and uses small models with modest computing requirements. We will compare different feature sets and models for year-round prediction, examining whether combining recent observations with historical seasonal information improves next-day mean-temperature forecasts.

## Dataset

### Data Source and Collection

We obtain ECMWF's ERA5 global reanalysis data (Hersbach et al., 2020) through the Open-Meteo Historical Weather API (Zippenfenig, 2023). ERA5 combines historical observations with numerical models, and Open-Meteo aggregates its hourly data into daily values. We wrote a Python script that calls the API through HTTP requests, converts the returned JSON into a table, and saves it as a CSV file.

We request data for Brown University at 41.83°N, 71.40°W. The returned ERA5 grid point is 41.75°N, 71.50°W. We explicitly select ERA5 at 0.25° resolution to maintain a consistent source from January 1, 2001, through December 31, 2025, with the timezone setting America/New_York.

### Dataset Size and Features

The modeling collection contains 9,131 daily records with 15 weather fields: 14 numerical variables and one categorical variable, weather_code. There are no missing dates, duplicate dates, or missing weather values. The fields include:

- Temperature: Daily mean, maximum, and minimum temperatures (°C).
- Atmospheric conditions: Relative humidity and cloud cover (%), and mean sea-level pressure (hPa).
- Precipitation: Total precipitation and rainfall (mm), snowfall (cm), and precipitation duration (hours).
- Wind and radiation: Maximum wind speed (km/h), dominant wind direction (°), shortwave radiation (MJ/m²), and daylight duration (seconds).
- Weather conditions: WMO weather codes, treated as categories.

The EDA figures use a separate team collection containing 6,941 daily records from January 1, 2006 through January 1, 2025. It also includes 19 weather fields, including surface pressure, wind gusts, reference evapotranspiration, and sunshine duration.

### Target and Input

The target is daily mean temperature on day d. Across the modeling collection, it has a mean of 11.4°C, a standard deviation of 9.4°C, and a range of −17.6°C to 32.1°C.

Inputs include the preceding 28 days and seven-day windows centered on the corresponding date in each of the previous five years, giving 63 historical positions. Each position contains 15 weather fields and two calendar encodings. Two additional encodings describe the target date, producing 1,073 raw predictors.

Complete historical windows yield 7,302 samples. We divide them chronologically by target date: training from January 4, 2006 through 2017, validation from 2018–2021, and testing from 2022–2025.

## Exploratory Data Analysis

We investigated seasonal weather patterns and relationships among meteorological features to predict weather conditions in Providence, Rhode Island. The analysis uses historical ERA5 reanalysis data obtained through the Open-Meteo Historical Weather API. We examined correlations among numerical variables, distributions of weather conditions, and long-term temperature and solar radiation patterns. These analyses help identify features associated with our prediction targets and evaluate whether seasonal patterns provide a useful foundation for weather prediction.

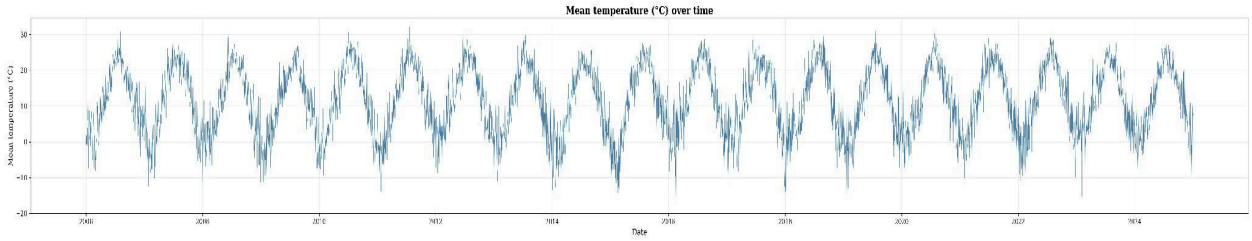
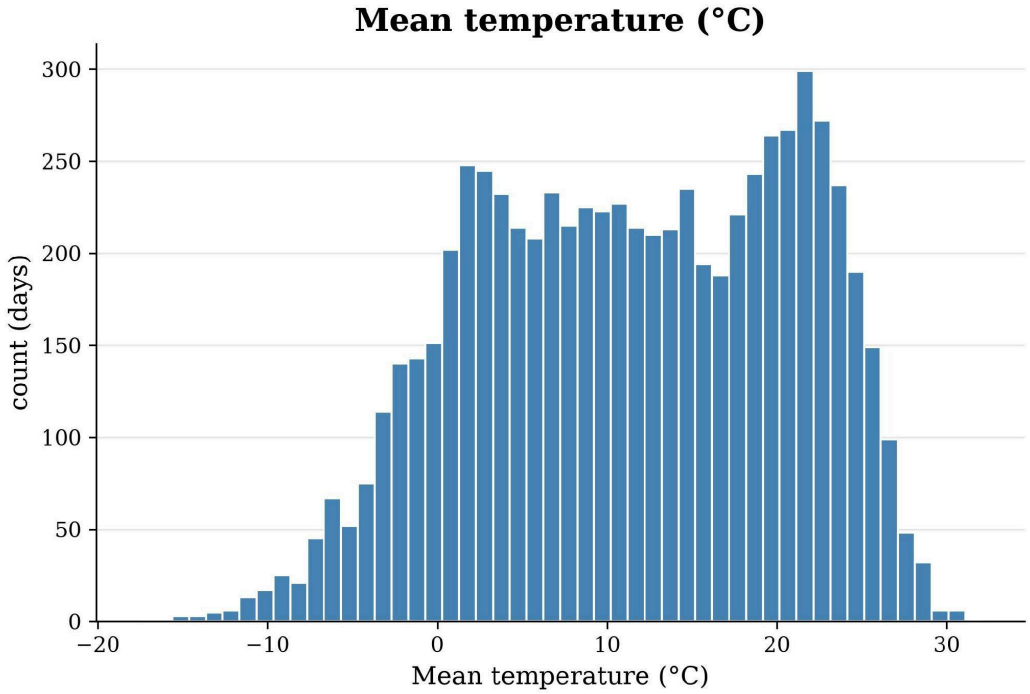



Figure 1. Daily mean temperature over the exploratory period.

Temperatures generally increase from winter to summer before declining toward the end of each year, with summer temperatures frequently reaching 25–30°C and winter temperatures occasionally falling below −10°C. However, the graph also shows some fluctuations within individual seasons and differences

between corresponding periods in different years. We know calendar features capture seasonal patterns, and recent weather may help predict the temperature on a specific day.




Daily mean air temperature at 2 m above ground, °C.

Figure 2. Distribution of daily mean temperature across all 6,941 days.

Daily mean temperatures range from approximately −17.6°C to 32.1°C. The pooled distribution contains observations from every season, with fewer days at the coldest and hottest extremes. Therefore, forecast errors should be examined across seasons and temperature ranges.

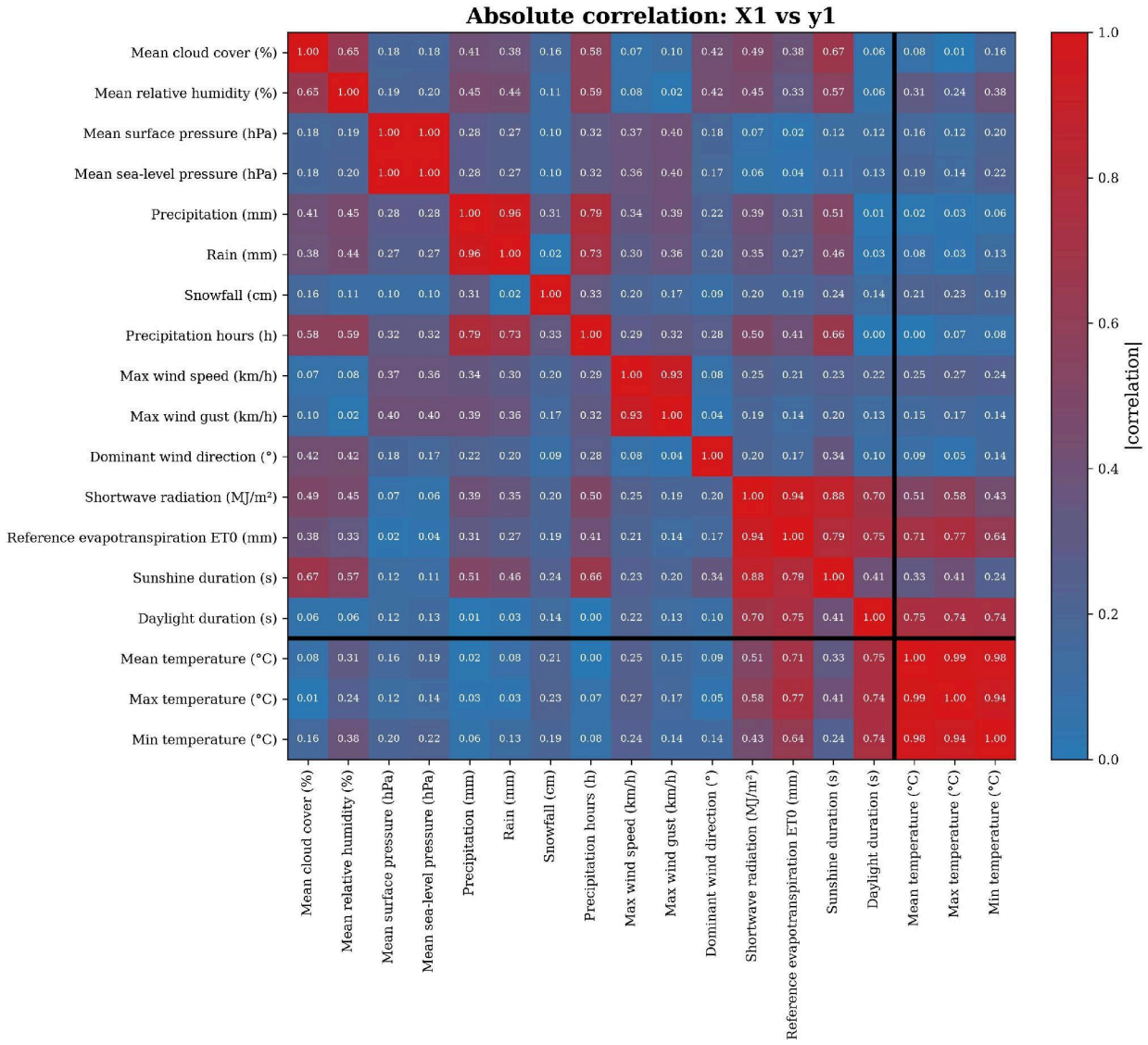



Figure 3. Absolute Pearson correlations among continuous daily weather measurements.

Daylight duration shows the strongest association with mean temperature among the non-temperature variables, with a correlation of 0.75, followed by reference evapotranspiration of 0.71 and shortwave radiation of 0.51. Mean temperature is also strongly correlated with maximum and minimum temperature (0.99 and 0.98). The matrix shows which variables are related to temperature and which contain similar information. These patterns can help us choose feature sets to compare in our forecasting models.

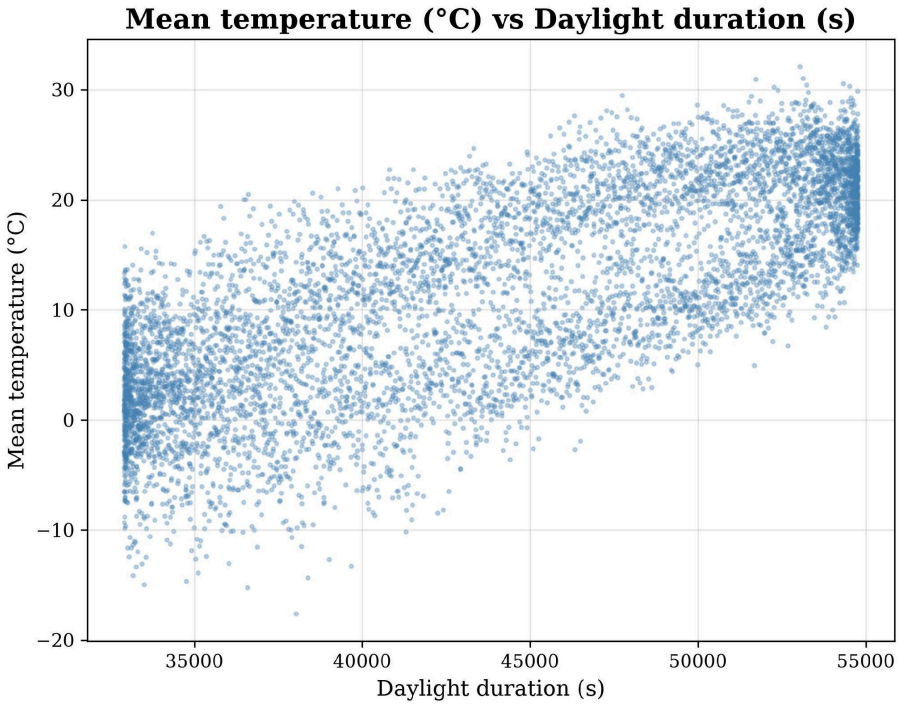



Figure 4. Daily mean temperature versus daylight duration, measured in seconds.

Longer days generally accompany higher temperatures, but similar daylight durations span a wide range of temperatures. This supports representing seasonality while leaving room for weather information beyond the calendar.

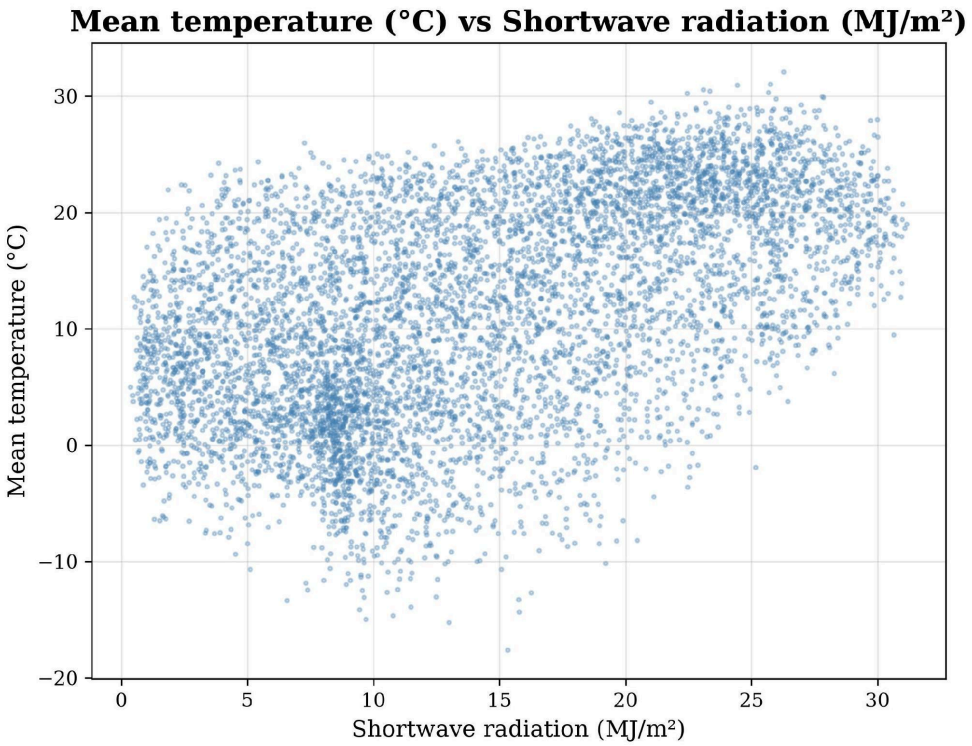



Figure 5. Daily mean temperature versus shortwave radiation.

Higher radiation is generally associated with warmer days, although temperatures vary substantially at similar radiation levels. Historical radiation is a candidate input; this plot does not show its contribution after accounting for season or other variables.

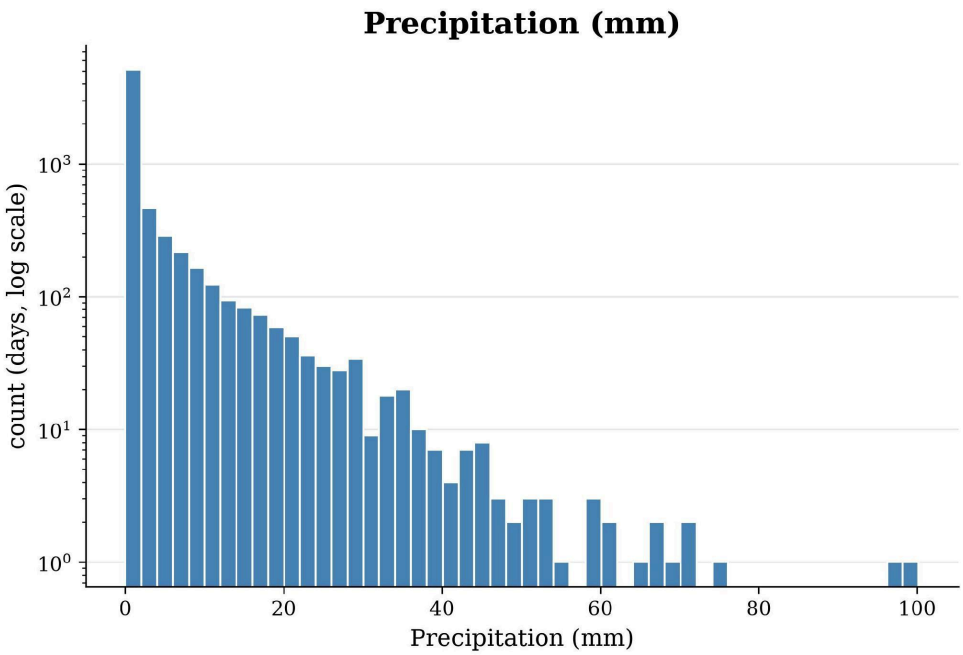



Figure 6. Distribution of total daily precipitation.

Precipitation is concentrated near zero with a long right tail. The CSV contains 3,643 days with zero precipitation (52.5%). Large precipitation values should be checked rather than removed simply because they are uncommon.

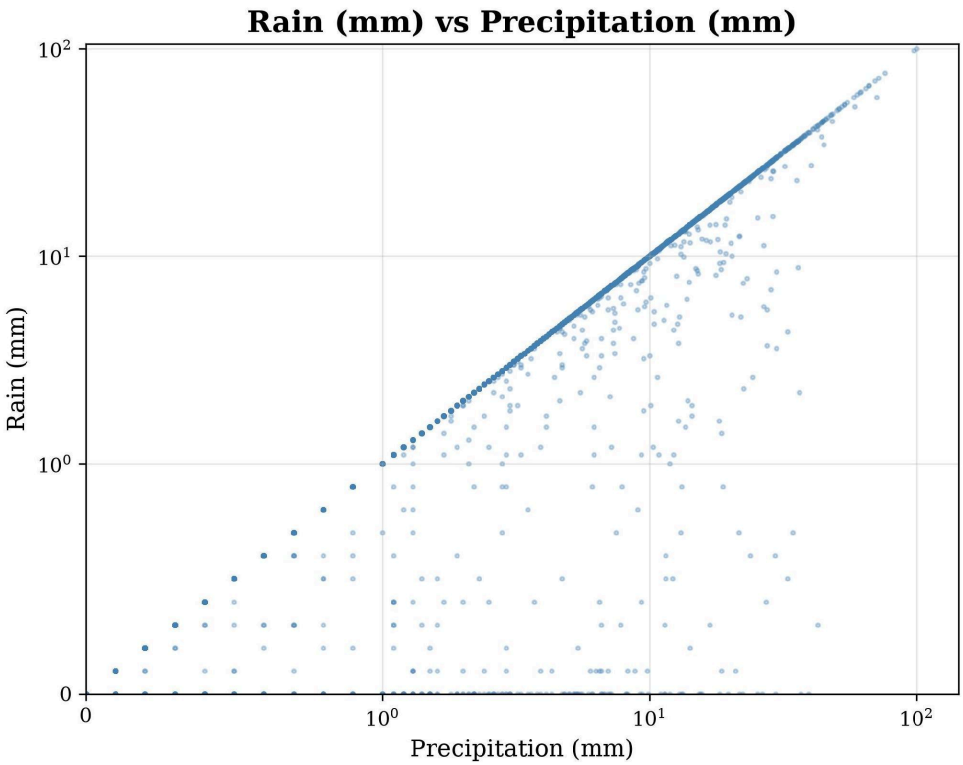



Figure 7. Rainfall versus total precipitation.

Total precipitation and rainfall have a correlation of 0.96, while precipitation and precipitation hours have a correlation of 0.79. This shows that precipitation-related measurements capture similar aspects of weather conditions. We know that rainfall is itself part of total precipitation. The highly correlated predictors can introduce redundancy, so further regularization and feature selection are important in later research.

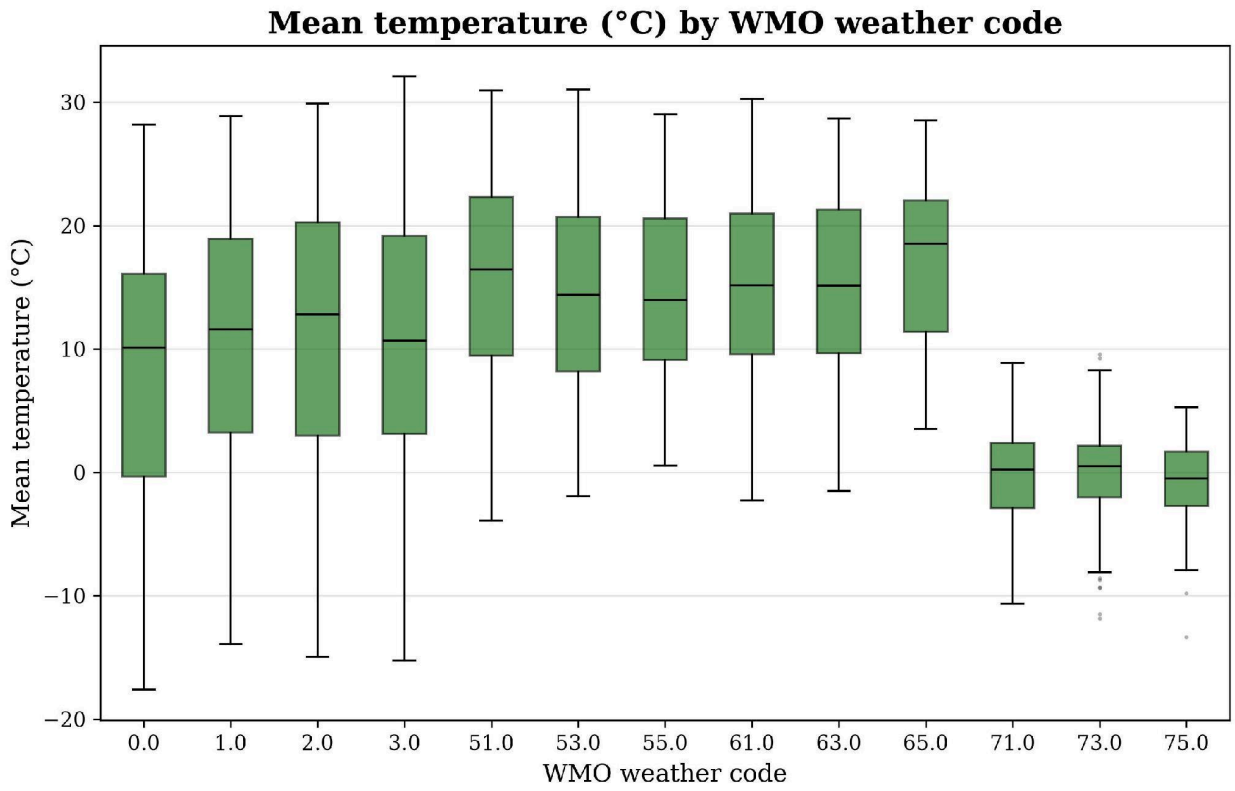



Figure 8. Daily mean temperature by WMO weather code.

Snow categories have lower temperatures, while clear, cloudy, and rainy categories overlap substantially. These differences also reflect season and how weather categories are defined. The weather codes 0–3 describe sky conditions, 51–55 describe drizzle, 61–65 describe rain, and 71–75 describe snow.

These figures together give us a preliminary picture of how temperature changes over time and how it relates to other weather variables. They also show that some features contain overlapping information, which we will further consider when selecting inputs. We will use these findings to compare different feature sets and build models for next-day mean-temperature prediction in our later work.

# Problem Description and Potential Impact

This project is a supervised regression task to predict the next-day mean temperature near Brown University. We will compare models using recent weather observations, historical seasonal information, and calendar features. We are interested in whether combining recent and seasonal information improves predictions compared with using either source alone. Comparing different input sets will also help us identify a simpler set of useful features.

We believe our potential users include Brown students, staff, and visitors, who could use temperature predictions to plan clothing, daily travel, and outdoor activities. The project will examine whether small models can provide useful local predictions with modest computing resources. We will also examine errors across seasons to understand when the models perform well and where improvements are needed. Our current experiments use historical ERA5 data. For live deployment, recent weather inputs would need to come from a timely observation source, since ERA5 is published with a delay.

# References

## Reference Work


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## Data Source


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